Lecture 7: Confidence intervals

Economics 326 — Methods of Empirical Research in Economics

Vadim Marmer, UBC

Point estimation

- Our model:
  1. $Y_i = \beta_0 + \beta_1 X_i + U_i, \qquad i = 1, \ldots, n.$
  2. $E(U_i | X_1, \ldots, X_n) = 0$ for all i's.
  3. $E(U_i^2 | X_1, \ldots, X_n) = \sigma^2$ for all i's.
  4. $E(U_i U_j | X_1, \ldots, X_n) = 0$ for all $i \neq j$.
  5. U's are jointly normally distributed conditional on X's.
- The OLS estimator $\hat{\beta}_1$ is a **point estimator** of β_1.
- For our model, conditional on X's:

$$\hat{\beta}_1 \sim N\left(\beta_1, Var\left(\hat{\beta}_1\right)\right),$$

$$Var\left(\hat{\beta}_1\right) = \frac{\sigma^2}{\sum_{i=1}^n \left(X_i - \bar{X}\right)^2}.$$

- With probability **one**, we have that $\hat{\beta}_1 \neq \beta_1$.

Interval estimation problem

- We want to construct an **interval estimator** for β_1:
  - The interval estimator is called a **confidence interval** (CI).
  - A CI contains the **true** value β_1 **with some pre-specified probability** $1 - \alpha$, where α is a small probability of error.
  - For example, if $\alpha = 0.05$, then the random CI will contain β_1 with probability 0.95.
- $1 - \alpha$ is called the **coverage probability**.
- Confidence interval: $CI_{1-\alpha} = [LB_{1-\alpha}, UB_{1-\alpha}]$. The lower bound (LB) and upper bound (UB) should depend on the coverage probability $1 - \alpha$.
- The formal definition of CI: It is a **random interval** $CI_{1-\alpha}$ such that conditional on X's,

$$P(\beta_1 \in CI_{1-\alpha}) = 1 - \alpha.$$

  Note that the random element is $CI_{1-\alpha}$.
- Sometimes, a CI is defined as $P(\beta_1 \in CI_{1-\alpha}) \geq 1 - \alpha$.

Symmetric CIs

- One approach to constructing CIs is to consider a **symmetric** interval around the estimator $\hat{\beta}_1$:

$$CI_{1-\alpha} = \left[\hat{\beta}_1 - c_{1-\alpha}, \hat{\beta}_1 + c_{1-\alpha}\right]$$

- The problem is choosing $c_{1-\alpha}$ such that $P\left(\beta_1 \in CI_{1-\alpha}\right) = 1-\alpha$.
- In choosing $c_{1-\alpha}$ we will be relying on the fact that given our assumptions and conditionally on X's:

$$\hat{\beta}_1 \sim N\left(\beta_1, Var\left(\hat{\beta}_1\right)\right) \text{ and } Var\left(\hat{\beta}_1\right) = \frac{\sigma^2}{\sum_{i=1}^n \left(X_i - \bar{X}\right)^2}.$$

- Note that conditionally on X's:

$$\frac{\hat{\beta}_1 - \beta_1}{\sqrt{Var\left(\hat{\beta}_1\right)}} \sim N\left(0,1\right).$$

Quantiles (percentiles) of the standard normal distribution

- Let $Z \sim N\left(0,1\right)$. The τ-th **quantile** (percentile) of the standard normal distribution is z_τ such that

$$P\left(Z \leq z_\tau\right) = \tau.$$

- **Median**: $\tau = 0.5$ and $z_{0.5} = 0$. ($P\left(Z \leq 0\right) = 0.5$).
- If $\tau = 0.975$ then $z_{0.975} = 1.96$. Due to symmetry, if $\tau = 0.025$ then $z_{0.025} = -1.96$.

σ^2 is known (infeasible CIs)

- **Suppose** (for a moment) that σ^2 is known, and we can compute exactly the variance of $\hat{\beta}_1$ as $Var\left(\hat{\beta}_1\right) = \sigma^2 / \sum_{i=1}^n \left(X_i - \bar{X}\right)^2$.
- Consider the following CI:

$$CI_{1-\alpha} = \left[\hat{\beta}_1 - z_{1-\alpha/2}\sqrt{Var\left(\hat{\beta}_1\right)}, \hat{\beta}_1 + z_{1-\alpha/2}\sqrt{Var\left(\hat{\beta}_1\right)}\right].$$

- For example, if $1-\alpha = 0.95 \iff \alpha = 0.05 \iff z_{1-\alpha/2} = z_{0.975} = 1.96$, and

$$CI_{0.95} = \left[\hat{\beta}_1 - 1.96\sqrt{Var\left(\hat{\beta}_1\right)}, \hat{\beta}_1 + 1.96\sqrt{Var\left(\hat{\beta}_1\right)}\right].$$

Validity of the infeasible CIs (σ^2 is known)

- We need to show that $P\left(\beta_1 \in \left[\hat{\beta}_1 - z_{1-\alpha/2}\sqrt{Var\left(\hat{\beta}_1\right)}, \hat{\beta}_1 + z_{1-\alpha/2}\sqrt{Var\left(\hat{\beta}_1\right)}\right]\right) = 1-\alpha$.
- Next,

$$\begin{aligned}
&\hat{\beta}_1 - z_{1-\alpha/2}\sqrt{Var\left(\hat{\beta}_1\right)} \leq \beta_1 \leq \hat{\beta}_1 + z_{1-\alpha/2}\sqrt{Var\left(\hat{\beta}_1\right)} \\
&\iff -z_{1-\alpha/2}\sqrt{Var\left(\hat{\beta}_1\right)} \leq \beta_1 - \hat{\beta}_1 \leq z_{1-\alpha/2}\sqrt{Var\left(\hat{\beta}_1\right)} \\
&\iff -z_{1-\alpha/2}\sqrt{Var\left(\hat{\beta}_1\right)} \leq \hat{\beta}_1 - \beta_1 \leq z_{1-\alpha/2}\sqrt{Var\left(\hat{\beta}_1\right)} \\
&\iff -z_{1-\alpha/2} \leq \frac{\hat{\beta}_1 - \beta_1}{\sqrt{Var\left(\hat{\beta}_1\right)}} \leq z_{1-\alpha/2}
\end{aligned}$$

Validity of the infeasible CIs (σ^2 is known)

- We have that

$$\begin{aligned}
&\beta_1 \in \left[\hat{\beta}_1 - z_{1-\alpha/2}\sqrt{Var\left(\hat{\beta}_1\right)}, \hat{\beta}_1 + z_{1-\alpha/2}\sqrt{Var\left(\hat{\beta}_1\right)}\right] \\
&\iff -z_{1-\alpha/2} \leq \frac{\hat{\beta}_1 - \beta_1}{\sqrt{Var\left(\hat{\beta}_1\right)}} \leq z_{1-\alpha/2}.
\end{aligned}$$

- Let $Z = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{Var\left(\hat{\beta}_1\right)}} \sim N(0,1)$.

$$\begin{aligned} &P\left(-z_{1-\alpha/2} \le \frac{\hat{\beta}_1 - \beta_1}{\sqrt{Var\left(\hat{\beta}_1\right)}} \le z_{1-\alpha/2}\right) \\ &= P\left(-z_{1-\alpha/2} \le Z \le z_{1-\alpha/2}\right) \\ &= P\left(z_{\alpha/2} \le Z \le z_{1-\alpha/2}\right) \\ &= 1 - \alpha/2 - \alpha/2 = 1 - \alpha. \end{aligned}$$

Feasible confidence intervals (σ^2 is unknown)

- Since σ^2 is unknown, we must estimate it from the data:

$$s^2 = \frac{1}{n-2} \sum_{i=1}^{n} \hat{U}_i^2 = \frac{1}{n-2} \sum_{i=1}^{n} \left(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i\right)^2.$$

- We can replace σ^2 by s^2, however, the result does not have a normal distribution any more:

$$\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\widehat{Var}\left(\hat{\beta}_1\right)}} \sim t_{n-2}, \text{ where } \widehat{Var}\left(\hat{\beta}_1\right) = \frac{s^2}{\sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2}.$$

  Here t_{n-2} denotes the t-distribution with $n-2$ degrees of freedom.

- The degrees of freedom depend on
  - the sample size (n),
  - and the number of parameters one have to estimate to compute s^2 (two in this case, β_0 and β_1).

Feasible confidence intervals (σ^2 is unknown)

- Let $t_{df,\tau}$ be the τ-th quantile of the t-distribution with the number of degrees of freedom df: If $T \sim t_{df}$ then

$$P\left(T \le t_{df,\tau}\right) = \tau.$$

- Similarly to the normal distribution, the t-distribution is centered at zero and is symmetric around zero: $t_{n-2,1-\alpha/2} = -t_{n-2,\alpha/2}$.
- We can now construct a feasible confidence interval with $1-\alpha$ coverage as:

$$CI_{1-\alpha} = \left[\hat{\beta}_1 - t_{n-2,1-\alpha/2}\sqrt{\widehat{Var}\left(\hat{\beta}_1\right)}, \hat{\beta}_1 + t_{n-2,1-\alpha/2}\sqrt{\widehat{Var}\left(\hat{\beta}_1\right)}\right],$$
$$\text{where } \widehat{Var}\left(\hat{\beta}_1\right) = \frac{s^2}{\sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2}.$$

Example: Rent rates and average income

- Data (RENTAL.DTA): 64 cities in 1990, Rent = average rent, AvgInc = per capita income: $\text{Rent}_i = \beta_0 + \beta_1 \text{AvgInc}_i + U_i$.
- $t_{62,0.975} \approx 2.00 \implies$ The **95%** confidence interval for β_1 is $[0.0115 - 2 \times 0.0013, 0.0115 + 2 \times 0.0013] = [0.0089, 0.0141]$.
- $t_{62,0.95} \approx 1.671 \implies$ The **90%** confidence interval for β_1 is $[0.0115 - 1.671 \times 0.0013, 0.0115 + 1.671 \times 0.0013] = [0.0093, 0.0137]$.

The effect of estimation of σ^2

- The t-distribution has heavier tails than normal.
- $t_{df,1-\alpha/2} > z_{1-\alpha/2}$, but as df increases $t_{df,1-\alpha/2} \to z_{1-\alpha/2}$.
- When the sample size n is large, $t_{n-2,1-\alpha/2}$ can be replaced with $z_{1-\alpha/2}$.

Interpretation of confidence intervals

- The confidence interval $CI_{1-\alpha}$ is a function of the **sample** $\{(Y_i, X_i) : i = 1, \ldots, n\}$, and therefore is **random**. This allows us to talk about probability of $CI_{1-\alpha}$ containing the true value of β_1.
- Once the confidence interval is computed given the data, we have its **one realization**. The realization of $CI_{1-\alpha}$ or (computed confidence interval) is not random, and it does not make sense anymore to talk about the probability that it includes the true β_1.
- **Once the confidence interval is computed, it either contains the true value β_1 or it does not**.